Vincent Wallstein

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EDUCATION

UNIVERSITY OF COLORADO BOULDER August 2025–May 2029
BS Mechanical Engineering
GPA: 3.7

SKILLS

SOLIDWORKS, AutoCAD, Onshape, MATLAB, 3D Printing, Minitab, Excel, Circuit Integration, Machining

EXPERIENCE

FORMULA SAE Boulder, Colorado
Member: Aerodynamics & Chassis Subteams September 2025–Present
- Designed and fabricated jigs and inserts to test carbon alternatives for push rods, tie rods, and toe links
- Assisted in developing the testing procedure and timeline for validation and implementation

STARLIGHT CARDIOVASCULAR San Diego, California
Intern: Nitinol Engineer May–July 2026
- Conducted regression analysis and statistical modeling on 65+ Nitinol stents to identify and quantify relationships between manufacturing parameters and mechanical performance; presented findings to company leadership
- Designed and proposed a 2-factor full factorial DOE to establish FDA-defensible manufacturing tolerances for pediatric cardiac stents; identified an orthogonality conflict due to manufacturing constraints, restructured the design accordingly

QUASAR MEDICAL Dongguan, China/Chonburi, Thailand
Intern: Mechanical Engineer Summer 2023, 2024
- Designed trimming jig for micro-tubing, ensuring a precise cut without clamping or stretching the material
- Developed and submitted tube-cutting solution for heart catheter automation using vacuum channel component; installed in machines producing 15,000+ units annually

LUCENT AUTO DETAILING Orange County, California
Founder & Owner March 2021–2024
- Built and managed mobile auto care business, handling marketing, client outreach/scheduling, and finances
- Trained team of three employees, scaled service capacity and increased client base by ~200%, servicing 40+ vehicles/year

PROJECTS

AUTONOMOUS SECURITY DRONE January 2026–Present
Personal Project
- Led mechanical and electrical development for an autonomous drone tailored for motion-triggered surveillance, real-time tracking, and user notification
- Designed drone frame and structural components in SolidWorks for 3D printing, developed electrical layout and wiring architecture integrating sensors, flight controller, and power systems.

ELECTROMAGNETIC AIRCRAFT LAUNCH SYSTEM - *SCALED MODEL* January–May 2026
Engineering Project (CU Boulder)
- Co-developed scaled electromagnetic aircraft launch system to reduce commercial aviation emissions during takeoff
- Collaborated with Navy engineers to adapt military EMALS technology for commercial applications
- Spearheaded design and prototyping of components, including coil assemblies and launch sleds using SolidWorks; integrated mechanical components with electrical circuit model
- Won Best in Section; received special recognition for ingenuity and resourcefulness from ThinkOrbital, an orbital defense and robotics company